Deep Learning Based Medical Image Analysis For Early Disease Detection

Deployed Application

Phase III project submission | 30 October 2026

Student / institution: enter the approved cover details. No identity or approval has been invented.

Deployment status

Public deployment URL: PENDING. No deployment platform account or credentials were available. The repository is prepared for deployment, but a public service has not been launched.

The working application was run locally at http://127.0.0.1:8511 during verification. This local address is not a public deployment URL. The source repository is private at https://github.com/PeterParker7810/deep-learning-medical-image-analysis.

Overview and verified functionality

Streamlit research prototype with Overview, Image analysis, Dataset status, Model results and Methodology sections. In the image workflow, an image is validated, previewed, preprocessed, passed to the selected trained checkpoint and shown with class probabilities. A Grad-CAM attention overlay is available for the verified ResNet-18 checkpoint; it is not a segmentation mask or clinical explanation.

The application has been exercised using a genuine image from the official ISIC test partition and the actual trained checkpoint. Dataset availability and metrics reflect generated local experiment artifacts.

Architecture and access

Run `python -m pip install -r requirements.txt`, place authorized dataset files in the documented local layout, run `python audit_data.py --task isic_skin_lesion --data-root data/raw/isic2016 --manifest metadata.csv`, then `streamlit run app.py`. The checkpoint is stored at `models/saved_models/isic_skin_lesion/resnet18/model.pt` locally. The app runs inference locally; image bytes are held in memory by the application code.

Limitations

NIH and APTOS pages show dataset-specific pending status. Public deployment is pending. The app is a research prototype, not clinically validated; predictions must not replace professional medical assessment. Avoid identifiable patient images.